郭麗蓮 113700044

10.3

17. Disinfectants An experiment was conducted to study the use of 95% ethanol versus 20% bleach as a disinfectant in removing contamination when culturing plant tissues. The experiment was repeated 15 times with each disinfectant, using cuttings of eggplant tissue.[9] The observation reported was the number of uncontaminated eggplant cuttings after a 4-week storage.

Disinfectant	95% Ethanol	20% Bleach
Mean	3.73	4.80
Variance	2.78095	.17143
n	15	15
	Pooled variance 1.47619	

a. Are you willing to assume that the underlying variances are equal?

b. Using the information from part a, is there evidence of a significant difference in the mean numbers of uncontaminated eggplants for the two disinfectants tested?

a) $\frac{\text{larger } s^2}{\text{smaller } s^2} = \frac{2.78095}{0.17143} \approx 16.22 > 3$

No, it not reasonable to assume variances are equal

b) ① $H_0: \mu_1 - \mu_2 = 0$

$H_a: \mu_1 - \mu_2 \neq 0$

② $\alpha = 0.05$ - two tailed

③ $t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \sim t_{df}$

$V = df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{(n_1-1)} + \frac{(s_2^2/n_2)^2}{(n_2-1)}} = \frac{\left(\frac{2.78095}{15} + \frac{0.17143}{15}\right)^2}{\frac{0.18546^2}{14} + \frac{0.01143^2}{14}} = \frac{0.09874}{0.002464} \approx 15$

④ $t = \frac{3.73 - 4.80}{\sqrt{\frac{2.78}{15} + \frac{0.17}{15}}} = \frac{-1.07}{\sqrt{0.19467}} = -2.41$

⑤ $t_{0.025,15} = 2.131 \rightarrow$ rejection region $= t < -2.131$ or $t > 2.131$

$-2.41 < -2.131$ (Reject H_0)

There is sufficient evidence that there is a significant difference in the mean numbers of uncontaminated eggplants for 2 disinfectants

20. Dissolved O_2 Content A pollution control inspector suspected that a river community was releasing amounts of semitreated sewage into a river. To check his theory, he drew five randomly selected specimens of river water at a location above the town, and another five below. The dissolved oxygen readings (in parts per million) are as follows:

Above Town	4.8	5.2	5.0	4.9	5.1
Below Town	5.0	4.7	4.9	4.8	4.9

a. Do the data provide sufficient evidence to indicate that the mean oxygen content below the town is less than the mean oxygen content above? Test using $\alpha = .05$.

b. Estimate the difference in the mean dissolved oxygen contents for locations above and below the town. Use a 95% confidence interval.

a) ① $H_0: \mu_B = \mu_A$

$H_a: \mu_B < \mu_A$

$\bar{x}_A = \frac{4.8+5.2+5.0+4.9+5.1}{5} = 5.00$

$\bar{x}_B = \frac{5.0+4.7+4.9+4.8+4.9}{5} = 4.86$

$5 - 4.86 = 0.14$

ABOVE

x	$(x-\bar{x})$	$(x-\bar{x})^2$
4.8	-0.2	0.04
5.2	0.2	0.04
5.0	0	0
4.9	-0.1	0.01
5.1	0.1	0.01
		0.10

$s_A^2 = \frac{\sum(x-\bar{x})^2}{n-1} = \frac{0.1}{4} = 0.025$

BELOW

x	$(x-\bar{x})$	$(x-\bar{x})^2$
5.0	0.14	0.0196
4.7	-0.16	0.0256
4.9	0.04	0.0016
4.8	-0.06	0.0036
4.9	0.04	0.0016
		0.052

$s_B^2 = \frac{\sum(x-\bar{x})^2}{n-1} = \frac{0.052}{4} = 0.013$

$s_p^2 = \frac{(n_A-1)s_A^2 + (n_B-1)s_B^2}{n_A+n_B-2} = \frac{4\times0.025 + 4\times0.013}{8} = \frac{0.152}{8} = 0.019$

$s_p = \sqrt{0.019} = 0.1378$

② $\alpha = 0.05$ - left tailed

③ $t_{stat} = \frac{\bar{x}_A - \bar{x}_B}{\sqrt{\frac{s_p^2}{n_A} + \frac{s_p^2}{n_B}}}$

④ $t_{stat} = \frac{5 - 4.86}{\sqrt{\frac{0.019+0.019}{5}}} = \frac{0.14}{\sqrt{\frac{0.019+0.019}{5}}} = \frac{0.14}{0.0871} = 1.607 = 1.61$, $df = 5+5-2 = 8$

⑤ $t_{0.05,8} = 1.860 \rightarrow$ rejection region $t < -1.860$

$1.61 > -1.860$ (Failed to reject H_0)

There is no enough evidence to conclude that oxygen below town is less than above town.

b) 95% Confidence Interval

$(\bar{x}_A - \bar{x}_B) \pm t_{\alpha/2,\, df} \cdot \sqrt{\frac{s_p^2}{n_A} + \frac{s_p^2}{n_B}}$ 0.0871

$0.14 \pm t_{0.025,8} \times 0.0871$

$0.14 \pm 2.306 \times 0.0871$

$0.14 \pm 0.201 \approx 0.201$ $(-0.061, 0.341)$

10.4

Paired-Difference Tests II *Use the information in Exercise 9 to calculate the observed value of the t statistic for testing the difference between the two population means using paired data. Find a rejection region with $\alpha = .05$ to state your conclusions. Construct a 95% confidence interval for $\mu_1 - \mu_2$. Does this confirm the results of your hypothesis test?*

9.

	Pairs				
Population	1	2	3	4	5
1	1.3	1.6	1.1	1.4	1.7
2	1.2	1.5	1.1	1.2	1.8
	0.1	0.1	0	0.2	-0.1

a) ① $H_0: \mu_D = 0$

$H_a: \mu_D \neq 0$

$\bar{d} = \frac{0.1+0.1+0+0.2-0.1}{5} = \frac{0.3}{5} = 0.06$

d	$d-\bar{d}$	$(d-\bar{d})^2$
0.1	0.04	0.0016
0.1	0.04	0.0016
0	-0.06	0.0036
0.2	0.14	0.0196
-0.1	-0.16	0.0256
		0.052

$s^2_d = \frac{0.052}{5-1} = 0.013$

$s_d = \sqrt{0.013} = 0.114$

② $\alpha = 0.05$ - two tailed

③ $t_{stat} = \frac{\bar{d}}{s_d/\sqrt{n}}$

④ $t_{stat} = \frac{0.06}{0.114/\sqrt{5}} = \frac{0.06}{0.051} = 1.17$, $df = 4$

⑤ $t_{0.025,4} = 2.776 \rightarrow$ rejection region $= t < -2.776$ or $t > 2.776$

$1.17 < 2.776$ (Failed to Reject H_0)

There is not enough evidence to conclude that the two population means are different

b) 95% Confidence Interval

$\bar{d} \pm t_{\alpha/2,(n-1)} \frac{s_d}{\sqrt{n}}$

$\bar{d} \pm t_{0.025,4} \times \frac{0.114}{\sqrt{5}}$

$0.06 \pm 2.776 \times 0.051$

0.06 ± 0.141 $(-0.081, 0.201)$

$\rightarrow$ Yes, the interval contains 0, so it confirms the hypothesis test

fail to reject H_0

DATA SET DS1026

20. Memory Experiments Twenty students participated in an experiment in which they were first asked to recall a list of 25 words. They were then given a second list, and asked to form images of the words as they were read. The number of words recalled for each student is given in the table.

Student	With Imagery	Without Imagery		d^2
1	20	5	+15	225
2	24	9	+15	225
3	20	5	+15	225
4	18	9	+9	81
5	22	6	+16	256
6	19	11	+8	64
7	20	8	+12	144
8	19	11	+8	64
9	17	7	+10	100
10	21	9	+12	144
11	17	8	+9	81
12	20	16	+4	16
13	20	10	+10	100
14	16	12	+4	16
15	24	7	+17	289
16	22	9	+13	169
17	25	21	+4	16
18	21	14	+7	49
19	19	12	+7	49
20	23	13	+10	100
				2415

$\Sigma d = 205 \rightarrow \bar{d} = \frac{205}{20} = 10.25$

Does it appear that the average recall score is higher when imagery is used?

$\Sigma d = 205 \rightarrow \bar{d} = \frac{205}{20} = 10.25$

$\Sigma d^2 = 2415$

$s^2_d = \frac{\Sigma d^2 - \frac{(\Sigma d)^2}{n}}{n-1} = \frac{2415 - \frac{205^2}{20}}{19} = \frac{2415 - 2101.25}{19} = 16.507 \approx 16.51$

$s_d = \sqrt{16.51} = 4.05$

① $H_0: \mu_D = 0$

$H_a: \mu_D > 0$

② $\alpha = 0.05$ - right tailed

③ $t_{stat} = \frac{\bar{d}}{s_d/\sqrt{n}}$

④ $t_{stat} = \frac{10.25}{4.05/\sqrt{20}} = \frac{10.25}{0.9056} = 11.318 \approx 11.32$, $df = 20-1 = 19$

⑤ $t_{0.025,19} = 1.729 \rightarrow$ rejection region $= t > 1.729$

$11.32 > 1.729$ (Reject H_0)

There is sufficient evidence to conclude that average recall is higher when imagery is used.

DATA SET DS1028

22. Cake Mixes An experiment was conducted to compare the densities (in ounces per cubic inch) of cakes prepared from two different cake mixes. Six cake pans were filled with batter A, and six were filled with batter B. Expecting a variation in oven temperature, the experimenter placed a pan filled with batter A and another with batter B *side by side* at six different locations in the oven. The six paired observations of densities are as follows:

Location	1	2	3	4	5	6
Batter A	.135	.102	.098	.141	.131	.144
Batter B	.129	.120	.112	.152	.135	.163

a. Do the data present sufficient evidence to indicate a difference between the average densities of cakes prepared using the two types of batter?

b. Construct a 95% confidence interval for the difference between the average densities for the two mixes.

a) ① $H_0: \mu_D = 0$

$H_a: \mu_D \neq 0$

A	B	d	d^2
0.135	0.129	0.006	0.000036
0.102	0.120	-0.018	0.000324
0.098	0.112	-0.014	0.000196
0.141	0.152	-0.011	0.000121
0.131	0.135	-0.004	0.000016
0.144	0.163	-0.019	0.000361
		-0.06	0.001054

$\Sigma d = -0.06$

$\bar{d} = \frac{-0.06}{6} = -0.010$

$S_d = \sqrt{\frac{\Sigma d^2 - \frac{(\Sigma d)^2}{n}}{n-1}} = \sqrt{\frac{0.001054 - \frac{(-0.06)^2}{6}}{5}} = \sqrt{\frac{0.001054 - 0.0006}{5}} = 0.0009528 = 0.00953$

$\bar{d} = \frac{-0.06}{6} = -0.01$

② $\alpha = 0.05$ - two tailed

③ $t_{stat} = \frac{\bar{d}}{S_d/\sqrt{n}}$

④ $t_{stat} = \frac{-0.01}{0.00953/\sqrt{6}} = \frac{-0.01}{0.00389} = -2.57$, $df = 6-1 = 5$

⑤ $t_{0.025,5} = 2.571 \rightarrow$ rejection region $= t < -2.571$ or $t > 2.571$

-2.57 is at the boundary (Reject H_0)

There is enough evidence to indicate a difference between the average densities

b) 95% confidence interval

$\bar{d} \pm t_{\alpha/2,(n-1)} \frac{S_d}{\sqrt{n}}$

$-0.01 \pm t_{0.025,5} \times 0.00389$

$-0.01 \pm 2.571 \times 0.00389$

-0.01 ± 0.01 $(-0.02, 0) \rightarrow$ The interval excludes positive values, confirming the hypothesis test.

10.5

DATA SET DS1031

12. A Precise Quarterback The number of passes completed and the total number of passing yards recorded for the Los Angeles Chargers quarterback, Philip Rivers for each of the 15 regular season games that he played in the fall of 2017[13] were used to calculate the average number of yards per pass for each game.

Week	Completions	Yards per Pass	Week	Completions	Yards per Pass
1	28	13.8	9	17	12.5
2	22	13.2	11	15	12.2
3	20	11.4	12	25	10.7
4	18	17.7	13	21	12.3
5	31	11.1	14	22	15.8
6	27	16.1	15	20	11.8
7	20	12.6	16	31	10.7
8	21	17.2	17	22	8.7

a. Calculate the mean and standard deviation of the number of completions and the number of yards per pass.

b. Find a 95% confidence interval estimate for the variance of the number of completions. Why would you prefer a small variability in the number of completed passes?

c. Find a 95% confidence interval for σ^2, the variance of the yards per pass. Use these results to find a 95% confidence interval for σ, the standard deviation of the yards per pass.

d. Test whether the standard deviation of the yards per pass for this quarterback differs from $\sigma = 4$ with $\alpha = .05$.

a) completions

$\Sigma x = 28+22+20+18+31+27+20+21+19+15+25+21+22+20+31+22$

$\Sigma x = 360$, $\Sigma x^2 = 8492$

mean $(\bar{x}) = \frac{360}{16} = 22.5$

$S^2 = \frac{\Sigma x^2 - \frac{(\Sigma x)^2}{n}}{n-1} = \frac{8492 - \frac{360^2}{16}}{15} = \frac{332}{15} = 22.13$

$s = \sqrt{22.13} = 4.704$

yards

$\Sigma x = 13.8+13.2+11.4+17.7+11.1+16.1+12.6+17.2+12.5+12.2+10.7+12.3+15.8+11.8+10.7+8.7$

$\Sigma x = 207.8$, $\Sigma x^2 = 2799.88$

mean $(\bar{x}) = \frac{207.8}{16} = 12.9875$

$S^2 = \frac{\Sigma x^2 - \frac{(\Sigma x)^2}{n}}{n-1} = \frac{2799.88 - \frac{(207.8)^2}{16}}{15} = \frac{96.079}{15} = 6.405 \approx 6.41$

$s = \sqrt{6.41} = 2.53$

b) 95% Confidence Interval for variance of completions

$\left[\frac{(n-1)s^2}{\chi^2_{(n-1),\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{(n-1),1-\alpha/2}}\right]$

$\left[\frac{15(22.13)}{\chi^2_{15,0.025}}, \frac{15(4.704)}{\chi^2_{15,0.975}}\right] = \left[\frac{332}{27.488}, \frac{332}{6.262}\right] = (12.08, 53.01)$

Small variability is preferred, because it means that quarterback is more consistent

c) 95% Confidence Interval for variance of yards per pass

$\left[\frac{(n-1)s^2}{\chi^2_{(n-1),\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{(n-1),1-\alpha/2}}\right]$

$\left[\frac{15(6.41)}{\chi^2_{15,0.025}}, \frac{15(6.41)}{\chi^2_{15,0.975}}\right] = \left[\frac{96.15}{27.488}, \frac{96.15}{6.262}\right] = (3.49, 15.354)$

95% Confidence Interval for standard deviation

$\sqrt{3.49} = 1.868$

$\sqrt{15.34} = 3.916$

$\Big\}$ $(1.868, 3.916)$

d) ① $H_0: \sigma = 4$

$H_a: \sigma \neq 4$

② $\alpha = 0.05$ - two tailed

③ $\chi^2_{stat} = \frac{(n-1)s^2}{\sigma^2_0} \sim \chi^2_{(n-1)}$

④ $\chi^2_{stat} = \frac{15(6.41)}{16} = \frac{96.15}{16} = 6.009375$

⑤ rejection region: $\chi^2 < 6.262$ or $\chi^2 > 27.488$

$6.009375 < 6.262$ (Reject H_0)

→ There is sufficient evidence that the standard deviation of yards / pass differs from 4

13. Instrument Precision A precision instrument is guaranteed to read accurately to within 2 units. A sample of four instrument readings on the same object yielded the measurements 353, 351, 351, and 355.

a. Test the null hypothesis that $\sigma = .7$ against the alternative $\sigma > .7$. Use $\alpha = .05$.

b. Find a 90% confidence interval for the population variance.

$\bar{x} = \frac{353+351+351+355}{4} = \frac{1410}{4} = 352.5$

x	$x-\bar{x}$	$(x-\bar{x})^2$
353	0.5	0.25
351	-1.5	2.25
351	-1.5	2.25
355	2.5	6.25
		11

$S^2 = \frac{11}{4-1} = \frac{11}{3} = 3.667$

$s = \sqrt{3.667} = 1.915$

a) ① $H_0: \sigma = 0.7$

$H_a: \sigma > 0.7$ $\rightarrow \sigma_0^2 = 0.49$

② $\alpha = 0.05$ - right tailed

③ $\chi^2_{stat} = \frac{(n-1)s^2}{\sigma_0^2}$

④ $\chi^2_{stat} = \frac{3(3.667)}{0.49} = 22.45$

⑤ $\chi^2_{0.05,3} = 7.815 \rightarrow$ rejection region $= \chi^2 > 7.815$

$22.45 > 7.815$ (Reject H_0)

There is sufficient evidence that the instrument variability is greater than 0.7 units

b) 90% Confidence Interval for Variance

$\left[\frac{(n-1)s^2}{\chi^2_{(n-1),\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{(n-1),1-\alpha/2}}\right]$

$\left[\frac{3(3.667)}{\chi^2_{3,0.05}}, \frac{3(3.667)}{\chi^2_{3,0.95}}\right] = \left[\frac{11}{7.815}, \frac{11}{0.352}\right] = (1.41, 31.25)$

The population variance is estimated to lie within 1.41 and 31.25